FINAL YEAR PROJECT

Project Title:

CareerCompass AI: An Intelligent Career Guidance and Placement Readiness System

Abstract:

The increasing competition in the job market has made it difficult for students to identify suitable career opportunities and understand the skills required for different job roles. Many students struggle to evaluate their strengths, identify skill gaps, and prepare effectively for placements.

The proposed system, "CareerCompass AI: Intelligent Career Guidance and Placement Readiness System," uses artificial intelligence to provide personalized career guidance and improve placement readiness. The system analyzes a student's educational background, technical skills, certifications, interests, and resume to recommend suitable career paths and job roles.

The platform also performs skill-gap analysis by comparing the student's current skills with industry requirements and generates a personalized learning roadmap. In addition, the system evaluates placement readiness, provides interview preparation assistance, and offers career-related guidance through an AI assistant.

The proposed solution aims to bridge the gap between academic learning and industry requirements while helping students make informed career decisions and improve their employment opportunities.

Problem Statement:

Students often face difficulties in selecting the right career path because they lack proper guidance, industry knowledge, and awareness of the skills required for different job roles. Existing platforms mainly focus on job searching rather than providing complete career guidance, skill-gap analysis, and placement preparation.

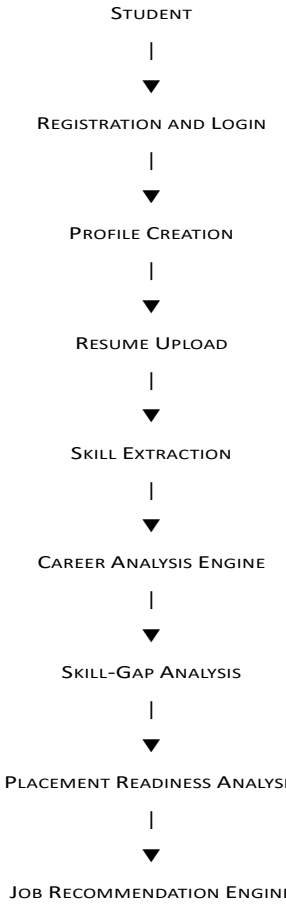
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This project aims to develop an artificial intelligence-based platform that analyzes students' profiles, identifies their strengths and weaknesses, recommends suitable career opportunities, and provides personalized guidance to improve their placement readiness.

Objective:

- Analyse students' resumes and profiles.
- Identify existing skills and missing skills.
- Recommend suitable career paths.
- Suggest appropriate job roles.
- Generate personalized learning roadmaps.
- Evaluate placement readiness.
- Provide interview preparation assistance.
- Offer career guidance using artificial intelligence.

System Architecture:



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MODULES:

MODULE 1 – AUTHENTICATION MODULE

- REGISTRATION
- LOGIN
- USER PROFILE MANAGEMENT

MODULE 2 – RESUME ANALYSIS MODULE

- RESUME UPLOADING
- PDF PROCESSING
- DATA EXTRACTION

MODULE 3 – SKILL ANALYSIS MODULE

- TECHNICAL SKILL IDENTIFICATION
- SOFT SKILL ANALYSIS
- CERTIFICATION ANALYSIS

MODULE 4 – CAREER RECOMMENDATION MODULE

- CAREER PREDICTION
- JOB ROLE SUGGESTIONS
- INDUSTRY RECOMMENDATIONS

MODULE 5 – SKILL-GAP ANALYSIS MODULE

- COMPARISON WITH INDUSTRY REQUIREMENTS
- MISSING SKILL IDENTIFICATION

- Improvement suggestions

Module 6 – Placement Readiness Module

- Aptitude assessment
- Communication assessment
- Overall readiness score

Module 7 – Learning Roadmap Module

- Daily goals
- Weekly goals
- Monthly goals

Module 8 – AI Assistant Module

- Interview preparation
- Career guidance
- Frequently asked questions

Technology Stack:

Components	Technologies
Front end	React.js
Back end	Node.js
Database	MongoDB
Artificial intelligence	Gemini API
Authentication	JWT
Styling	Tailwind CSS

Expected Outcome:

Your system should be able to answer questions such as:

- Which career path is suitable for me?
- Which skills should I learn?
- How ready am I for placements?
- Which companies should I apply to?
- What is the best learning path for me?

Expected Results:

- Career path recommendations
- Job suggestions
- Placement readiness score
- Skill-gap analysis report
- Personalized learning roadmap

Conclusion:

CareerCompass AI is an intelligent system designed to help students make better career decisions. The system analyses students' skills, interests, educational backgrounds, certifications, and resumes to recommend suitable career paths and job opportunities.

It also identifies skill gaps, evaluates placement readiness, and provides a personalized learning roadmap to help students improve their knowledge and skills. By combining artificial intelligence with career guidance, the system helps students prepare for their future in a more efficient and organized manner.

Overall, this project aims to reduce confusion among students, improve career planning, and increase employment opportunities.